

PUNTA DEL ESTE, Uruguay

WMO: 865860

Lat: 34.860S

Lon: 55.097W

Elev: 115

StdP: 14.64

Time Zone: -3.00 (W03)

Period: 00-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	36.1	39.0	31.6	26.0	44.6	32.1	26.6	44.9	29.5	49.0	25.9	50.6	3.9	300	0.504

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	17.5	89.5	71.6	86.4	70.5	84.1	69.8	75.2	83.4	73.7	81.2	72.4	79.4	11.8	40

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
73.2	123.9	78.7	71.6	117.0	77.1	70.0	111.0	75.6	38.9	83.1	37.4	80.0	36.2	79.7	84.6

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 26.8	23.1	20.8	DB	31.8	96.7	2.3	2.3	30.2	98.3	28.9	99.7	27.6	100.9	26.0	102.6	(4)
(5)			WB	31.6	79.9	2.1	2.1	30.1	81.4	28.9	82.6	27.7	83.8	26.2	85.4	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	62.2	73.1	72.4	68.9	63.7	57.9	53.0	51.6	54.1	56.2	60.8	65.3	69.8	(6)	
	DBStd	9.27	4.93	4.79	4.83	5.57	5.59	5.59	6.26	6.55	5.85	5.41	5.81	5.62	(7)	
	HDD50	121	0	0	0	0	4	26	53	27	10	1	0	0	(8)	
	HDD65	1983	3	3	19	89	229	365	418	346	271	154	65	21	(9)	
	CDD50	4565	716	628	587	410	249	115	103	153	196	335	458	615	(10)	
	CDD65	954	254	211	140	49	9	4	3	8	8	23	74	171	(11)	
	CDH74	5890	1789	1257	625	158	30	14	13	60	51	155	514	1224	(12)	
	CDH80	1693	579	365	122	22	2	1	1	16	12	29	142	402	(13)	
(14) Wind	WSAvg	10.1	10.4	10.4	9.8	9.3	8.8	9.2	9.5	10.1	11.1	10.9	10.9	11.1	(14)	
(15) Precipitation	PrecAvg	45.4	3.3	4.1	4.4	4.4	3.7	4.3	4.0	3.1	3.3	4.3	3.8	3.2	(15)	
	PrecMax	61.7	5.8	9.5	15.7	9.5	10.2	8.5	7.2	6.2	6.3	10.4	7.0	8.2	(16)	
	PrecMin	27.7	0.7	1.0	0.7	0.2	0.2	1.7	1.0	0.6	0.6	0.7	1.1	0.6	(17)	
	PrecStd	9.0	1.4	2.4	3.5	2.4	2.1	1.9	1.9	1.6	1.5	2.5	1.7	2.2	(18)	
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	93.8	92.8	87.7	82.7	78.2	75.5	75.4	82.3	80.4	83.9	89.3	91.6	(19)	
		MCWB	72.5	72.9	72.0	69.9	65.9	67.4	64.9	67.9	67.9	67.4	69.2	71.3	(20)	
	2%	DB	89.4	87.7	83.6	78.4	72.3	69.4	68.2	73.1	73.3	78.5	84.2	87.9	(21)	
		MCWB	72.1	72.3	70.1	66.6	63.2	62.9	61.0	62.3	61.7	65.1	67.6	69.5	(22)	
	5%	DB	86.0	84.4	80.4	75.2	69.0	64.6	63.9	67.9	69.5	73.8	80.2	84.4	(23)	
		MCWB	71.0	71.4	69.4	66.3	62.1	59.9	58.7	60.6	59.9	62.8	66.4	68.7	(24)	
	10%	DB	82.7	80.9	77.3	72.2	66.2	61.2	60.7	64.1	66.0	70.2	75.6	80.9	(25)	
		MCWB	70.1	70.2	68.4	65.7	61.1	56.5	56.5	58.1	58.6	61.6	64.5	67.7	(26)	
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	77.9	76.9	75.5	72.7	68.3	68.2	66.4	67.6	69.3	71.1	72.7	76.2	(27)	
		MCDB	85.9	86.4	82.2	78.4	74.0	74.0	72.8	78.6	77.2	79.0	83.5	86.4	(28)	
	2%	WB	75.3	74.8	73.2	70.4	65.7	64.4	62.4	64.3	65.1	67.5	70.0	72.9	(29)	
		MCDB	83.7	82.7	78.6	74.6	69.7	67.8	67.0	70.7	70.4	73.7	78.5	81.2	(30)	
	5%	WB	73.5	73.4	71.6	68.9	63.7	61.1	59.8	61.8	62.1	65.3	68.1	71.1	(31)	
		MCDB	81.0	80.9	76.9	72.6	67.2	63.9	63.0	66.4	66.3	70.7	76.0	79.2	(32)	
	10%	WB	72.1	71.9	70.0	67.2	61.9	58.0	57.4	59.3	59.9	63.5	66.3	69.5	(33)	
		MCDB	79.2	78.5	75.0	71.0	65.3	60.6	60.2	63.0	64.2	68.7	73.2	77.0	(34)	
(35) Mean Daily Temperature Range	5% DB	MDBR	17.5	15.7	15.3	14.9	14.2	14.5	14.1	14.8	15.0	15.6	17.5	17.7	(35)	
		MCDBR	22.6	20.2	19.1	18.3	17.2	17.7	17.7	21.3	21.9	22.2	23.7	23.7	(36)	
	5% WB	MCWBR	11.0	9.9	9.8	10.0	10.5	12.1	11.8	12.9	12.4	12.0	11.8	11.3	(37)	
		MCWBR	18.4	17.1	14.6	14.0	13.5	15.1	15.5	18.2	17.2	17.7	19.4	18.6	(38)	
(40) Clear-Sky Solar Irradiance	taub		0.392	0.391	0.370	0.365	0.355	0.362	0.361	0.443	0.454	0.396	0.381	0.384	(40)	
		taud	2.430	2.453	2.500	2.462	2.464	2.421	2.397	2.106	2.098	2.336	2.408	2.424	(41)	
	Ebn at Noon		299	293	287	269	253	240	247	237	255	288	301	303	(42)	
		Edh at Noon	39	36	32	30	26	26	28	42	48	41	39	39	(43)	
(44) All-Sky Solar Radiation	RadAvg		2308	1975	1647	1200	833	701	732	968	1342	1751	2165	2364	(44)	
	RadStd		163	182	141	130	56	64	69	95	98	151	171	154	(45)	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (5 neighbors)		+0.86	+1.26	N/A	+1.24	+1.13	N/A	N/A	N/A	+225	+121

Nomenclature: See separate page